AMERICAN ACADEMY OF OPHTHALMOLOGY®

Developing Laterality-Specific Computable Phenotypes from Electronic Health Record Data, Employing Treatment-Warranted Diabetic Macular Edema as a Use Case

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Purpose: To develop a general algorithm employing structured and unstructured electronic health record (EHR) data to identify laterality-specific treatment-warranted disease more accurately at the longitudinal eye level.

Design: A retrospective treatment-warranted diabetic macular edema (TW-DME) cohort study.

Subjects: Patients with diabetic retinopathy (DR) identified from a health safety net system and a university hospital in Los Angeles, California, employing diagnosis and procedure codes from 2013 to 2023.

Methods: We investigated the completeness and accuracy of laterality-specific TW-DME status based on the following 5 categories of data: Tier 1-Physician Procedure Documentation, Tier 2-Charge Codes (Professional and Facility), Tier 3-Medication Orders, Tier 4-Crosswalked Procedure Codes, and Tier 5-Diagnosis Code associated with Procedure. Laterality data completeness was evaluated for each category, independently and in a tiered hierarchical order. Data accuracy was verified by manual chart review for a subset of validation patients.

Main Outcome Measures: Algorithm performance in ascertaining cross-sectional and longitudinal TW-DME status.

Results: From 2013 to 2023, 7784 patients with DR had 68 465 visits, with 4809 (61.8%) patients identified as having TW-DME. Notably, 67.9% of health safety net patients had visits with missing diagnosis laterality. The proposed algorithm improved laterality completeness in the treatment-warranted DR cohort to 93.6% for the safety net and 99.0% for the university sites. Validation by chart review demonstrated an increase in positive predictive value (safety net 47.0%–93.2%, university 85.3%–98.8%), negative predictive value (safety net 23.2%–33.3%, university 46.9%–72.6%), sensitivity (safety net 35.9%–76.0%, university 79.2%–96.0%), specificity (safety net 60.4%–76.6%, university 38.8%–90.4%), agreement (safety net 38.5%–76.1%, university 74.8%–95.4%), and F1 score (safety net 40.7%–83.7%, university 82.1%–97.4%) at the longitudinal eye level.

Conclusions: Our algorithm employing structured and unstructured data lays out a general and reproducible approach to more accurately identify and extract laterality-specific data from EHRs. This method was valid across sites with disparate documentation and coding practices. Application of this algorithm could improve the utility of clinical data generated as part of routine care for future investigations of ocular disease prevalence, sequelae, treatment patterns, and costs.

Financial Disclosure(s): Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article. *Ophthalmology Science 2025;5:100797 © 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Supplemental material available at www.ophthalmologyscience.org.

Electronic health records (EHRs) and medical claims[1–3] provide an appealing source of big data collected as part of routine care[4–6] that has become prevalent in clinical research. Various repositories of EHR data in ophthalmology[7–10] are available and are increasingly being employed to study the development, progression, and treatment response of diseases[11–14] such as glaucoma, age-related macular degeneration, and diabetic retinopathy (DR).[15–18]

Current literature on the validity of coded ocular disease data primarily focuses on patient-level data at a single visit, with limited attention to longitudinal eye-level data. Findings from previous studies have shown varying concordance metrics between billing and medical record documentation in EHRs ranging from 0.61 to 1, and primarily focused on the patient level.[1,2,4,5,8,9,11,15,16,19] With regard to treatment-warranted diabetic macular edema (TW-DME), there have been various strategies employed to iteratively improve the utility of available data to develop an accurate computable definition. Bearelly et al[2] conducted a study involving 225 patients and demonstrated that an algorithm based on International Classification of Diseases, Ninth Revision (ICD-9) codes could accurately detect patient-level diabetic

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https://doi.org/10.1016/j.xops.2025.100797
ISSN 2666-9145/25

macular edema (DME) in billing data, achieving a sensitivity of 0.88 and a specificity of 0.96. Cai et al[11] subsequently reported that ICD, Tenth Revision (ICD-10) coding is more accurate than ICD-9, particularly for proliferative DR (PDR), showing improved accuracy over time. Lau et al[8] examined the accuracy of billing codes used in the therapeutic care of DR. Borkar et al[20] showed that integrating supplemental billing codes for therapeutic and diagnostic procedures with standard diagnosis codes can improve the accuracy of diagnoses in large EHRs. Limited data are available on the validity of computable phenotypes at the longitudinal eye level, which is critical for understanding the full trajectory of ocular conditions, including their progression and response to treatment over time.

A potential limitation of EHRs when conducting ocular research is the impact of missing or erroneous data. In addition, a significant portion of EHR data is likely to be unstructured (e.g., free text), which might be overlooked in systems heavily relying on billing codes, such as ICD diagnosis codes, during data extraction. This issue may be exacerbated among clinical practice settings where the coded data do not underlie reimbursement, including capitated, safety net, and other systems.

Complicating widespread use of EHR data for ocular research is the lack of universal adoption of data standards for ophthalmic record coding among EHR vendors. For example, the National Health Service in the United Kingdom utilizes the "DR structured assessment module" from the Medisoft EHR system to calculate the grade of DR and maculopathy,[7] while the Sight Outcomes Research Collaborative Ophthalmology Data Repository uses data from the Epic EHR system, which includes information collected as part of its Kaleidoscope data entry format.[10] Efforts have been made to standardize ocular data aggregation by the American Academy of Ophthalmology's Intelligent Research in Sight (IRIS) Registry,[21–23] but the process is highly manual and must be repeated for each EHR source contributing to the database.

We present an algorithm using both structured and unstructured ophthalmic clinical and ophthalmic EHR data to more accurately identify and extract treatment data at the longitudinal eye level. Using TW-DME as a test case, we assessed how well the algorithm could identify an ocular condition and associated treatment compared with manual chart review and standard cohort identification methods employing administrative data. Accurate database information plays a crucial role in assessing the disease burden and treatment effectiveness for diabetes mellitus and its ophthalmic complications.

Methods

Study Population

This was a methodological validation study employing multicenter retrospective, observational, longitudinal EHR data. The study population consisted of large samples of patients with DME in diverse clinical settings. The Los Angeles County Department of Health Services is one of the largest health safety net systems and public hospitals in the United States, serving a high proportion of low-income Latino patients, the largest minority group in the United States with a disproportionate burden of DME. The Los Angeles County Department of Health Services EHR contains data from 400 000 individuals across Southern California, collected from 2 EHR systems: Affinity (Harris Affinity) and Cerner PowerChart (Oracle Cerner), and aggregated into the Los Angeles County Department of Health Services data warehouse from 2013 to 2023. The study population also included the patient population from the University of Southern California (USC) Roski Eye Institute, which is a tertiary academic referral center. The USC Roski EHR database, sourced from Cerner PowerChart and aggregated into the USC Data Warehouse, includes data from 100 000 individuals who received care from 2020 to 2023.

The study (HS-21-00758) was approved by the USC Institutional Review Board. Informed consent was not required because the study used preexisting deidentified EHR data and met the exemption criteria under 45 Code of Federal Regulation 46.104(d)(4), posing no more than minimal risk to patients' privacy. All aspects of the research adhered to the Health Insurance Portability and Accountability Act and the principles outlined in the Declaration of Helsinki.

Patient Identification

Treatment-Warranted DR Cohort. We screened an initial treatment-warranted DR (TW-DR) cohort employing diagnosis codes for any DR (ICD-9: 362.0* or ICD-10: E08.3*, E09.3*, E10.3*, E11.3*, E12.3*, E13.3*), combined with focal laser (Current Procedural Terminology [CPT] code 67210) or intravitreal injection (CPT code 67028, accompanied by Healthcare Common Procedure Coding System procedure code for bevacizumab (J3490, J3590, J9035, Q2024, C9257, J7999), ranibizumab (J2778, C9233, C9093, Q5124), or aflibercept (J0178) at the same visit.

Treatment-Warranted Type II DME Cohort. Structured data were programmatically extracted for further review to select a cohort of TW-DME from the TW-DR cohort based on any of the following diagnoses codes: ICD-9: 362.07, 362.53, 362.83, with a previous diagnosis of type 2 diabetes (250.xx) or ICD-10: E11.311, E11.351, E11.37, E11.321, E11.331, E11.341. The exclusion criteria consisted of patients without a DME diagnosis at any visit, age <30 years old at baseline, patients without a diagnosis of type II diabetes, and those with a diagnosis of wet age-related macular degeneration and central retinal vein occlusion (Fig 1).

Longitudinal Chart Review Validation Cohort. Manual chart reviews of physician note texts from 316 patients validated the algorithm's accuracy in classifying treatment-warranted type II DME in the correct eye at each visit, with results compared with the coded data.

Algorithm Development: Visit-Level Laterality Determination. Data regarding laterality of eye disease and treatment were abstracted from 5 potential sources into 1 of 4 categories: right eye, left eye, both or bilateral eyes, or unknown. The algorithm used a regular expression to identify specific patterns or characters within a string. Note titles were stored in a field separate from the note body. Unstructured data such as procedure note titles were typically short phrases. Search terms to account for variations in terminology and abbreviations are listed in Table S1 (available at www.ophthalmologyscience.org). Laterality information from each source was evaluated for completeness and accuracy within the TW-DR cohort to determine the hierarchy in which to employ the data to assign the laterality for a given visit. Tier 1 employed abstraction from the free-text titles of physician procedure documentation. Tier 2 determined laterality from CPT or Healthcare Common Procedure Coding System professional and facility charge code modifiers (-RT/-LT/-50). Tier 3 derived

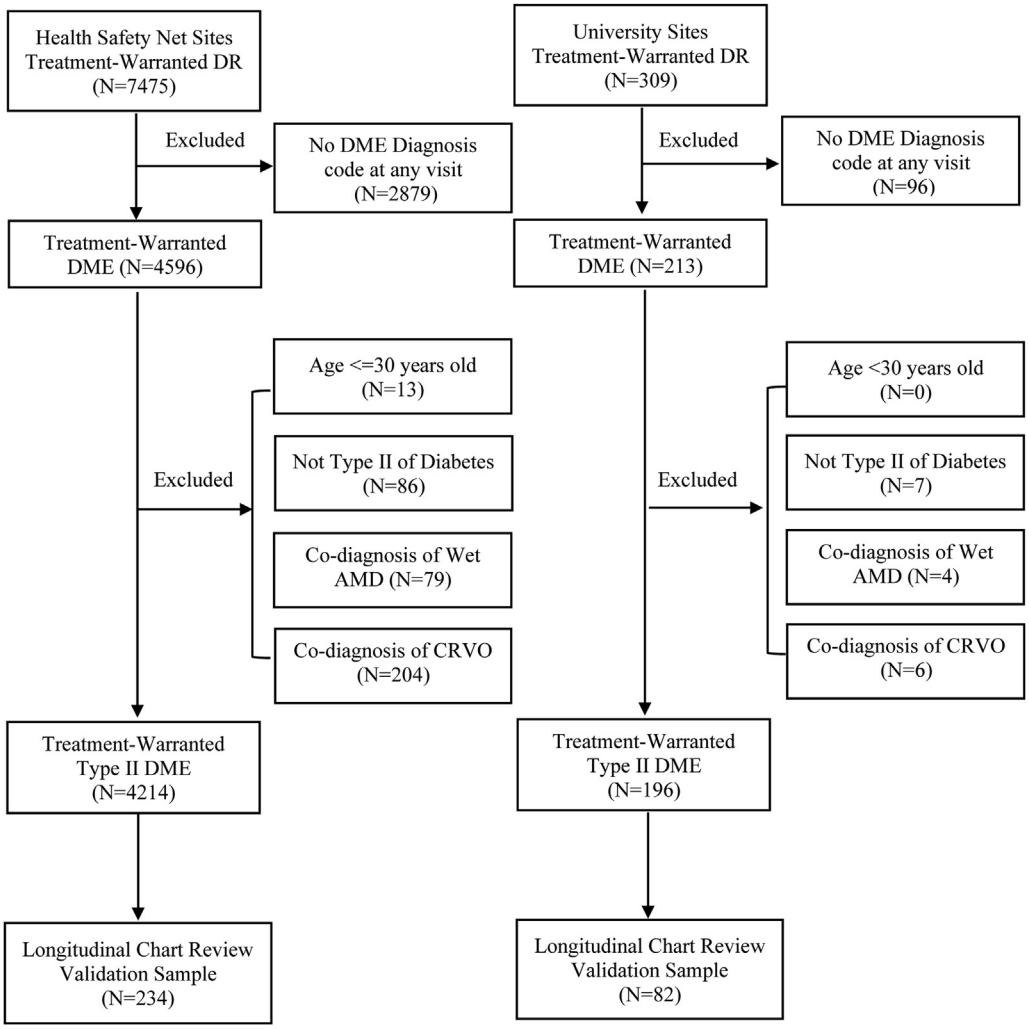


Figure 1. Attrition flow-diagram for study population inclusion and exclusion criteria. AMD = age-related macular degeneration; CRVO = central retinal vein occlusion; DME = diabetic macular edema; DR = diabetic retinopathy; N = number of unique patients.

laterality from the discrete field specifying the route of administration (right eye, left eye, or both eyes) from medication orders. Tier 4 leveraged ICD-Procedure Coding System procedure codes that intrinsically specify laterality and were manually entered by professional coders. Tier 5 extracted laterality from the descending rank-ordered ICD-9/ICD-10 diagnosis codes associated with each procedure. Figure 2 illustrates the proposed algorithm for assigning laterality in TW-DME using structured and unstructured EHR data across multiple tiers using a hierarchical series analysis. Initially, longitudinal eye-level laterality was assigned based on Tier 1 information. If laterality remained unknown in Tier 1, analysis progressed to Tier 2A and Tier 2B. If laterality still could not be determined, analysis advanced to subsequent tiers, including Tiers 3, 4, and 5. Finally, the bilateral modifiers were adjusted to encompass both left and right eyes, and duplicates were eliminated.

Algorithm Development: Employing the Longitudinal Record to Impute Missing Data. Imputation of missing data was based on the following 2 assumptions. First, when a person is diagnosed with PDR, the diagnosis persists even after treatment. We believe this assumption is reasonable, given that eyes with PDR are likely to remain affected within a 2-year period.[23] Our second assumption was that in a person who receives any 3 consecutive intravitreal injections with a time interval of 21 to 70 days between injections, with concordant first and third injection TW-DME laterality but missing second injection, TW-DME laterality should be assigned the same DME diagnosis for the intervening visit. Since DME is likely to persist for months even with intravitreal injection, we found this to be a reasonable assumption to make.[12,24]

Analysis

Data Completeness: Laterality Ascertainment by Algorithm Tiers. For the entire TW-DR cohort, the 5 tiers of laterality information were examined to determine the percentage of laterality data completeness, both independently and in series. Laterality information for each tier was independently reported as "RT-right," "LT-left," "BL-bilateral," or "Unknown" by number and percentage (Table 2). A hierarchical series analysis was conducted and percent of eyes with laterality (RT, LT, BL, or unknown) identified for the first time at each tier was reported, moving in order from Tier 1 to Tier 5 (Table 3).

Data Accuracy: Agreement between Algorithm-Assigned Laterality and Manual Chart Review. To validate the longitudinal accuracy of the algorithm, 2 authors (T.Z.L. and K.D.) independently reviewed the full text of clinical notes from each eye clinic visit for all 316 patients in the validation cohort, focusing on

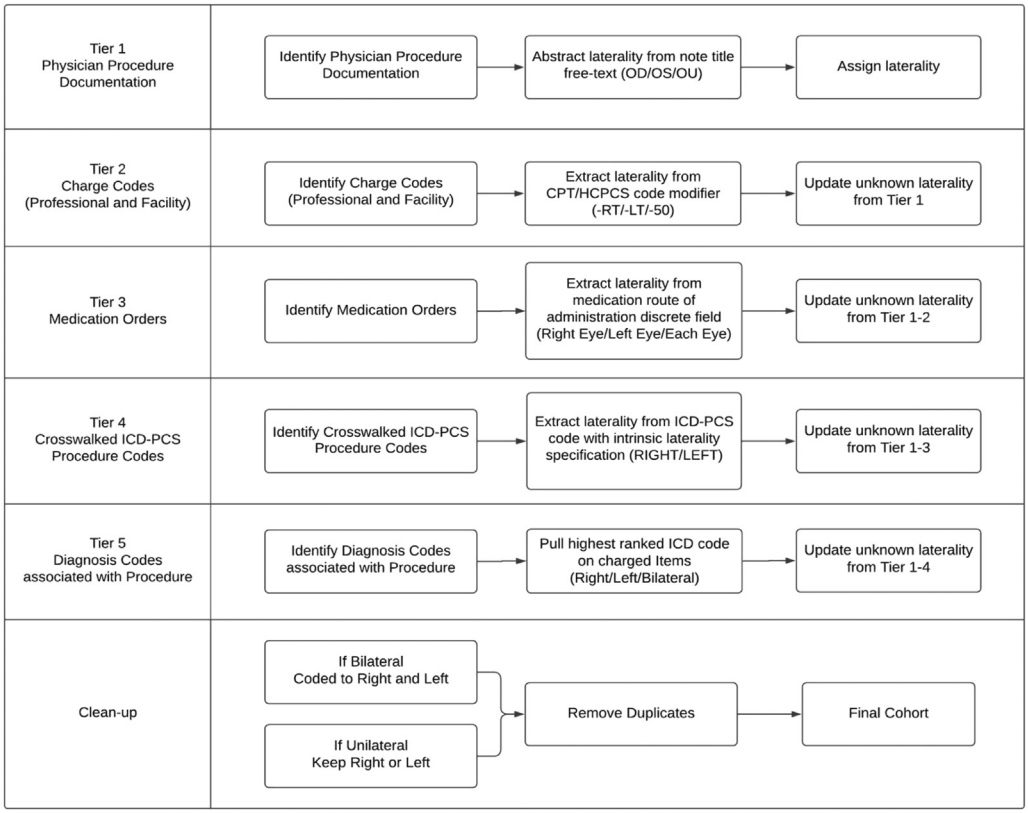


Figure 2. Flow-diagram of the proposed algorithm for assigning TW-DME treatment laterality using structured and unstructured data. CPT = Current Procedural Terminology; HCPCS = Healthcare Common Procedure Coding System; ICD = International Classification of Diseases; LT = left; RT = right; OD = right eye; OS = left eye; OU = both eyes; PCS = Procedure Coding System; TW-DME = treatment-warranted diabetic macular edema.

the presence of DR, DME, any associated treatments, and corresponding laterality. This reference standard was compared against the full algorithm, as well as subsets of the algorithm, with Tier 2 (CPT code) + Tier 5 (ICD code) corresponding to standard practice for code-based cohort identification.

We evaluated TW-DME algorithm performance at 3 levels: (1) eye-level at the index visit, which referred to accurately determining which eye (left, right, or both) was affected by DME at the first visit and received anti-VEGF treatment; (2) longitudinal patient-level, which referred to accurately ascertaining DME status at the person level (≥1 eye had TW-DME) for every visit; and (3) longitudinal eye-level analysis monitored each eye individually over time and referred to accurately determining TW-DME status for each eye at every visit.

The positive predictive value (PPV), negative predictive value (NPV), sensitivity, specificity, agreement, and F1 score were calculated at the eye-level for the index visit, as well as at the longitudinal patient-level and the longitudinal eye-level, to ascertain the validity of each method for identifying various cohorts. Data cleaning and analysis was conducted using RStudio (PBC, Version 1.4.1717, 2009-2022, available at: http://r-project.org).

Results

The TW-DR cohort comprised 7475 patients from the health care safety net sites and 309 patients from the university sites. The subset of patients who were included in the TW-DME cohort comprised 4596 (61.5%) patients within the health safety net sites and 213 (68.9%) patients within the university sites. The longitudinal chart review validation cohort comprised a random sample of 234 patients from the health safety net sites and 82 patients from the university sites (Fig 1).

TW-DME Cohort Characteristics

Patients with TW-DME were younger in the health safety net sites compared with the university sites, aged 62.8 ± 9.5 versus 66.4 ± 12.5 years (mean ± standard deviation, $P < 0.01$). The sex distribution was similar between the 2 TW-DME cohorts, with 58.4% male patients at the health safety net sites and 61.5% at the university sites ($P = 0.48$). The health care safety net sites had more self-identified Hispanic patients (81.3% vs. 27.7%, $P < 0.01$), fewer White patients (3.6% vs. 29.1%, $P < 0.01$), more patients with PDR (68.4% vs. 55.9%, $P < 0.01$), and greater utilization of bevacizumab at all visits (87.6% vs. 37.1%, $P < 0.01$). Full details are available in Table 4.

Data Completeness: Laterality Ascertainment by Algorithm Tiers

We evaluated the completeness of each tier of laterality data (Table 2). For the health safety net TW-DR cohort, the total coverage of laterality was as follows: 75.2% for Tier 1-Physician Procedure Documentation, 49.2% for Tier 2A-Charge Codes (Professional), 20.0% for Tier 2B-Charge

Table 2. Data Completeness: Laterality Ascertainment by Each Tier Independently

		Health Safety Net Sites (7475 Patients)		University Sites (309 Patients)	
		2013–2023		*2020–2023*	
		65 605 Visit Records	*Independent*	*2860 Visit Records*	*Independent*
Description	**Laterality**	*Total N (%)*	*N (%)*	*Total N (%)*	*N (%)*
Tier 1	RT	49 308 (75.2)	24 196 (36.9)	2598 (90.8)	1122 (39.2)
Physician Procedure Documentation	LT		24 550 (37.4)		1108 (38.7)
	BL		562 (0.9)		368 (12.9)
	Unknown		16 297 (24.8)		262 (9.2)
Tier 2A	RT	32 290 (49.2)	15 654 (23.9)	2792 (97.6)	1224 (42.8)
Charge Codes (Professional)	LT		15 275 (23.3)		1201 (42.0)
	BL		1361 (2.1)		367 (12.8)
	Unknown		33 315 (50.8)		68 (2.4)
Tier 2B	RT	13 103 (20.0)	5764 (8.8)		
Charge Codes (Facility)	LT		5679 (8.7)		
	BL		1660 (2.5)		
	Unknown		53 442 (81.5)		
Tier 3	RT	10 923 (16.6)	5176 (7.9)	0 (0.0)	0 (0.0)
Medication Orders	LT		5327 (8.1)		0 (0.0)
	BL		420 (0.6)		0 (0.0)
	Unknown		54 682 (83.4)		0 (0.0)
Tier 4	RT	0 (0.0)	0 (0.0)	58 (2.0)	29 (1.0)
Crosswalked ICD-PCS Procedure Codes	LT		0 (0.0)		29 (1.0)
	BL		0 (0.0)		0 (0.0)
	Unknown		0 (0.0)		1338 (97.6)
Tier 5	RT	21 092 (32.1)	3182 (4.9)	1619 (56.6)	349 (12.2)
Diagnosis Code associated with Procedure	LT		3106 (4.7)		223 (7.8)
	BL		14 804 (22.6)		1047 (36.6)
	Unknown		44 513 (67.9)		1241 (43.4)

BL = both eyes; Charge Codes (Facility) = procedure laterality derived from Current Procedural Terminology/Healthcare Common Procedure Coding System facility fee charge code modifier; Charge Codes (Professional) = procedure laterality derived from Current Procedural Terminology professional fee charge code modifier; Crosswalked ICD-PCS Procedure Codes = procedure laterality derived from International Classification of Diseases-Procedure Coding System charge code with intrinsic laterality specification; Diagnosis Code associated with Procedure = procedure laterality derived from International Classification of Diseases, Ninth Revision/International Classification of Diseases, Tenth Revision diagnosis code associated with the procedure; ICD = International Classification of Diseases; Independent N (%) = independent frequency count and percentage of right eye, left eye, and both eyes of each tier's laterality information; LT = left eye; PCS = Procedure Coding System; RT = right eye; Total N (%) = the total count visits with laterality information and coverage, reported as number and percentage of total visit records; Unknown = unspecified, unknown, or missing laterality.
Notes: Health Safety Net Sites: Los Angeles County Department of Health Services, Los Angeles General Medical Center; University Sites: University of Southern California Roski Eye Institute.

Codes (Facility), 16.6% for Tier 3-Medication Orders, and 32.1% for Tier 5-Diagnosis Code associated with Procedure. For the university TW-DR cohort, the total coverage of laterality was 90.8% for Tier 1, 97.6% for Tier 2, 0% for Tier 3, 2.0% for Tier 4, and 56.6% for Tier 5. After applying the proposed tiered algorithm with imputation, the health safety net sites saw a significant increase in the completeness of longitudinal eye-level laterality to 93.6%. Similarly, the university sites also experienced a substantial improvement, reaching 99.0% completeness for longitudinal eye-level laterality (Table 3).

Data Accuracy: Agreement between Algorithm-Assigned Laterality and Manual Chart Review

The overall agreement between programmatically extracted nonmissing laterality from different tiers and manual chart was high. For the health safety net sites validation cohort, the agreement rates were 97.8% for Tier 1-Physician Procedure Documentation, 97.5% for Tier 2A-Charge Codes (Professional), 95.7% for Tier 2B-Charge Codes (Facility), 90.7% for Tier 3-Medication Orders, and 50.6% for Tier 5-Diagnosis Code associated with Procedure. This implied that when the laterality information was available, there tended to be strong agreement. However, the agreement was lower when including all unknown laterality data from each tier as the denominator: 70.0% for Tier 1, 72.7% for Tier 2A, 18.1% for Tier 2B, 28.3% for Tier 3, and 25.7% for Tier 5 (Table 5). The lower agreement rates in this analysis were primarily due to the high percentage of missing laterality information, particularly for Tier 2B (81.1%) and Tier 5 (49.2%). In the university sites validation cohort, the overall agreement between programmatically extracted laterality data and manual chart review was high: 99.5% for Tier 1 and 98.9% for Tier 2 (Table 6). This decrease was only slightly affected by the inclusion of unknown laterality, as the university site has more complete data overall.

Algorithm Performance to Ascertain Cross-Sectional and Longitudinal TW-DME Status

Implementing the tiered algorithm with imputation to ascertain TW-DME status improved performance metrics including PPV, NPV, sensitivity, specificity, agreement, and

Table 3. Data Completeness: Laterality Ascertainment by Algorithm Tiers

		Health Safety Net Sites (7475 Patients)		University Sites (309 Patients)	
		2013–2023		*2020–2023*	
		65 605 Visit Records	*Series*	*2860 Visit Records*	*Series*
Description	**Laterality**	*Unknown N (%)*	*N (%)*	*Unknown N (%)*	*N (%)*
Tier 1	RT	49 308 (75.2)	24 196 (36.9)	2598 (90.8)	1122 (39.2)
Physician Procedure Documentation	LT		24 550 (37.4)		1108 (38.7)
	BL		562 (0.9)		368 (12.9)
	Unknown		16 297 (24.8)		262 (9.2)
Tier 2A	RT	16 297 (24.8)	4360 (6.7)	262 (9.2)	115 (4.0)
Charge Codes (Professional)	LT		4148 (6.3)		91 (3.2)
	BL		1106 (1.7)		22 (0.8)
	Unknown		6683 (10.2)		34 (1.2)
Tier 2B	RT	6683 (10.2)	190 (0.3)		
Charge Codes (Facility)	LT		194 (0.3)		
	BL		377 (0.6)		
	Unknown		5922 (9.0)		
Tier 3	RT	5922 (9.0)	243 (0.4)	34 (1.2)	0 (0.0)
Medication Orders	LT		249 (0.4)		0 (0.0)
	BL		35 (0.1)		0 (0.0)
	Unknown		5395 (8.2)		0 (0.0)
Tier 4	RT	5395 (8.2)	0 (0.0)	34 (1.2)	1 (0.0)
Crosswalked ICD-PCS Procedure Codes	LT		0 (0.0)		2 (0.0)
	BL		0 (0.0)		0 (0.0)
	Unknown		5395 (8.2)		31 (1.1)
Tier 5	RT	5395 (8.2)	125 (0.2)	31 (1.1)	0 (0.0)
Diagnosis Code associated with Procedure	LT		117 (0.2)		0 (0.0)
	BL		962 (1.5)		1 (0.0)
	Unknown		4191 (6.4)		30 (1.0)
Overall Completeness			93.60%		99.00%

BL = both eyes; Charge Codes (Facility) = procedure laterality derived from Current Procedural Terminology/Healthcare Common Procedure Coding System facility fee charge code modifier; Charge Codes (Professional) = procedure laterality derived from Current Procedural Terminology professional fee charge code modifier; Crosswalked ICD-PCS Procedure Codes = procedure laterality derived from International Classification of Diseases-Procedure Coding System charge code with intrinsic laterality specification; Diagnosis Code associated with Procedure = procedure laterality derived from International Classification of Diseases, Ninth Revision/International Classification of Diseases, Tenth Revision diagnosis code associated with the procedure; ICD = International Classification of Diseases; LT = left eye; PCS = Procedure Coding System; RT = right eye; Series N (%) = a hierarchical series analysis was conducted and percent of eyes with laterality (right eye, left eye, both eyes, Unknown) identified for the first time at each tier was reported, moving in order from Tier 1 to Tier 5; Unknown = unspecified, unknown, or missing laterality; Unknown N (%) = the total count of the visit with unknown laterality information after assigning right eye, left eye, and both eyes from the previous tier, reported as number and percentage of total visit records.
Notes: Health Safety Net Sites: Los Angeles County Department of Health Services, Los Angeles General Medical Center; University Sites: University of Southern California Roski Eye Institute.

F1 Score, for both sites, cross-sectionally and longitudinally, at the patient and eye levels. The performance improvement was most prominent for the longitudinal eye level. The PPV rose from 47.0% to 93.2% for safety net and from 85.3% to 98.8% for university; NPV also improved, with safety net increasing from 23.2% to 33.3% and university from 46.9% to 72.6%; sensitivity climbed from 35.9% to 76.0% in safety net settings and from 79.2% to 96.0% in university settings; specificity showed similar trends, rising from 60.4% to 76.6% for safety net and from 38.8% to 90.4% for university; agreement increased from 38.5% to 76.1% in safety net settings and from 74.8% to 95.4% in university settings; lastly, the F1 score reflected these enhancements, with safety net improving from 40.7% to 83.7% and university from 82.1% to 97.4% (Table 7). The individual breakdown for each site at different levels is available in Tables S8 to S12 (available at www.ophthalmologyscience.org).

Discussion

Our algorithm employing structured and unstructured EHR data in a tiered fashion and employing imputation of missing laterality was found to identify DME treatment data accurately and reliably at the longitudinal eye level in both the county-wide integrated health system and private academic hospital settings.

Most of the previous publications on research employing database or EHR data or medical records, as outlined in Tables S13 and S14 (available at www.ophthalmologyscience.org), primarily evaluated data accuracy at the patient level using ICD-9/ICD-10 and CPT or Healthcare Common Procedure Coding System codes. For example, 1 study utilizing the ICD-9 codes for patient-level DME diagnosis found 86% sensitivity and 84% specificity.[2] The study did not systematically evaluate the completeness of coverage at

Table 4. Characteristics of the Initial Treatment-Warranted DR Population by Site and DME Status

Study Site	Health Safety Net Sites			University Sites			
Time	*2013–2023*			*2020–2023*			
Patients Characteristics	*DME*	*No DME*	*Total*	*DME*	*No DME*	*Total*	
Number of Patients	*(N = 4596)*	*(N = 2879)*	*(N = 7475)*	*(N = 213)*	*(N = 96)*	*(N = 309)*	***P-Value**
Row %	61.5	38.5	100	68.9	31.1	100	
Patient level							
Baseline age (yrs), mean (SD)	62.8 (9.5)	60.2 (10.3)	61.8 (9.9)	66.4 (12.5)	69.1 (15.9)	67.2 (13.7)	<0.001
Male (N, %)	2685 (58.4)	1697 (58.9)	4382 (58.6)	131 (61.5)	55 (57.3)	186 (60.2)	0.62
Race (N, %)							<0.001
American Indian/Alaska Native	93 (2.0)	55 (1.9)	148 (2.0)	0 (0.0)	0 (0.0)	0 (0.0)	
Asian or Asian American	185 (4.0)	109 (3.8)	294 (3.9)	21 (9.9)	7 (7.4)	28 (9.1)	
Black or African American	173 (3.8)	92 (3.2)	265 (3.5)	12 (5.6)	4 (4.2)	16 (5.2)	
Native Hawaiian/Pacific Islander	13 (0.3)	8 (0.3)	21 (0.3)	2 (0.9)	0 (0.0)	2 (0.6)	
Hispanic or Latino	3735 (81.3)	2368 (82.3)	6103 (81.6)	59 (27.7)	22 (22.9)	81 (26.2)	
White	167 (3.6)	103 (3.6)	270 (3.6)	62 (29.1)	28 (29.2)	90 (29.1)	
Other/mixed	2 (0.0)	0 (0.0)	2 (0.0)	55 (25.8)	31 (32.3)	86 (27.8)	
Unknown	228 (5.0)	144 (5.0)	372 (5.0)	2 (0.9)	4 (4.2)	6 (1.9)	
Patient level, longitudinal							
DR stage (N, %)							<0.001
Mild NPDR	463 (10.1)	98 (3.4)	561 (7.5)	71 (33.3)	20 (11.9)	91 (29.4)	
Moderate NPDR	610 (13.3)	132 (4.6)	742 (9.9)	56 (26.3)	8 (4.8)	64 (20.7)	
Severe NPDR	1092 (23.8)	214 (7.4)	1306 (17.5)	53 (24.9)	2 (1.2)	55 (17.8)	
Stable NPDR	139 (3.0)	106 (3.7)	245 (3.3)	3 (1.4)	0 (0.0)	3 (1.0)	
PDR	3144 (68.4)	2342 (81.3)	5486 (73.4)	119 (55.9)	27 (16.1)	146 (47.2)	
Drug type (N, %)							<0.001
Bevacizumab	4028 (87.6)	2217 (77.0)	6245 (83.5)	79 (37.1)	18 (18.8)	97 (31.4)	
Aflibercept	235 (5.1)	16 (0.6)	251 (3.4)	96 (45.1)	18 (18.8)	114 (36.9)	
Ranibizumab	-	-	-	18 (8.5)	2 (2.1)	20 (6.5)	
Corticosteroid	412 (9.0)	32 (1.1)	444 (5.9)	22 (10.3)	2 (2.1)	24 (7.8)	
Unknown	875 (19.0)	444 (15.4)	1319 (17.6)	12 (5.6)	56 (58.3)	68 (22.0)	

DME = diabetic macular edema; DR = diabetic retinopathy; NPDR = nonproliferative diabetic retinopathy; Other/mixed = other race groups including >1 race; PDR = proliferative diabetic retinopathy; SD = standard deviation.
Notes: Health Safety Net Sites: Los Angeles County Department of Health Services, Los Angeles General Medical Center; University Sites: University of Southern California Roski Eye Instute.
Counts for DR stage and drug type do not sum to total due to varying number of visits and laterality of eye involvement among longitudinal sample. Corticosteroid: Dexamethasone, Iluvien, Kenalog, Ozurdex, Subtenon, Triesence, Triamcinolone, Methotrexate, and Fluocinolone. Continuous variable was displayed as mean (SD). Categorical variables were displayed as number N (% proportion).
**P* values for continuous and categorical variables were calculated from *t* test and Pearson chi-square test, respectively. *P* values were based on 2-sided comparisons of totals across study sites, with values <0.05 considered statistically significant.

the longitudinal eye level. Another study utilizing CPT and drug codes (J codes) for DR treatment found a PPV and NPV exceeding 89% and 88%, respectively.[8] However, the study only included 146 insured patients in retina clinics. Longitudinal data are vital for identifying risk factors, tracking disease progression, and assessing treatment effectiveness and advancement over time. Other larger databases, such as the American Academy of Ophthalmology's IRIS Registry, indicate that 53.0% to 89.5% of patient data, depending on the eye condition, do not contain eye-level laterality information.[21,22]

A significant challenge with EHR data is that approximately 80.0% remains unstructured and therefore underutilized, despite its substantial potential value. Authors of papers in other specialties have proposed the integration of structured and unstructured data from EHRs to develop clinically defined patient cohorts. To our knowledge, this is the first paper to combine structured and unstructured EHR data for a large volume of ophthalmology patients with DME. Systematic evaluations were conducted at the longitudinal eye level across disparate practice settings and diverse patient characteristics. The algorithm demonstrated a substantial improvement in completeness and accuracy of validity measures for a large EHR cohort when compared with current standard methods relying exclusively on codified data, especially at the longitudinal eye level. It increased laterality completeness to 93.6% in the health care safety net TW-DR cohort and 99.0% in the university TW-DR cohort. Additionally, it improved various performance metrics across both validation cohorts. Overall, the F1 score at the longitudinal eye level increased from 40.7% to 83.7% in the health care safety net validation cohort and from 82.1% to 97.4% in the university validation cohort.

The level of coding accuracy found at each medical center could be influenced by various factors such as the

Table 5. Data Accuracy: Agreement between Algorithm-Assigned Laterality and Manual Chart Review at Health Safety Net Sites

	Manual Chart Review					Algorithm Tier 2 Charge Codes					
	Assessment and Plan					*(Professional and Facility)*					
	(234 Patients)					**(7475 Patients)**					
Laterality	*RT*	*LT*	*Unknown*	*Agreement*[a] *(%)*	*Agreement*[b] *(%)*	*RT*	*LT*	*BL*	*Unknown*	*Agreement*[a] *(%)*	*Agreement*[b] *(%)*
Tier 1-Physician Procedure Documentation											
RT	621	10	0			12 812	181	46	11 025		
LT	19	656	0	97.8	70.0	198	12 721	80	11 359	98.1	40.1
BL	21	20	0			12	7	151	379		
Unknown	272	262	3			4420	4231	930	5567		
Tier 2A-Charge Codes (Professional)											
RT	624	19	0								
LT	15	628	1	97.5	72.7						
BL	46	44	0								
Unknown	229	237	3								
Tier 2B-Charge Codes (Facility)											
RT	151	8	0								
LT	7	127	0	95.7	18.1						
BL	28	26	0								
Unknown	723	763	4								
Tier 3-Medication Orders											
RT	229	28	0			2675	55	209	1862		
LT	27	293	0	90.7	28.3	72	2823	212	1839	91.3	8.9
BL	7	7	0			96	95	72	138		
Unknown	673	625	4			14 634	14 189	913	24 504		
Tier 5-Diagnosis Code associated with Procedure											
RT	234	14	0			2065	177	8	957		
LT	23	250	0	50.6	25.7	148	1988	4	982	26.1	7.1
BL	232	204	2			6333	6112	460	1963		
Unknown	444	479	4			8851	8820	709	24 263		

BL = both eyes; Charge Codes (Facility) = procedure laterality derived from Current Procedural Terminology/Healthcare Common Procedure Coding System facility fee charge code modifier; Charge Codes (Professional) = procedure laterality derived from Current Procedural Terminology professional fee charge code modifier; Crosswalked ICD-PCS Procedure Codes = procedure laterality derived from International Classification of Diseases-Procedure Coding System charge code with intrinsic laterality specification; Diagnosis Code associated with Procedure = procedure laterality derived from International Classification of Diseases, Ninth Revision/International Classification of Diseases, Tenth Revision diagnosis code associated with the procedure; LT = left eye; RT = right eye; Unknown = unspecified, unknown, or missing laterality.

Notes: Health Safety Net Sites: Los Angeles County Department of Health Services, Los Angeles General Medical Center.

One visit of the same patient could have multiple records. Agreement as % was calculated as the total number of cases with laterality (including all RT, LT, and BL) in agreement divided by the total number of cases.

Agreement[a] denominator was total eyes without unknown laterality.

Agreement[b] denominator was total eyes (including unknown).

Table 6. Data Accuracy: Agreement between Algorithm-Assigned Laterality by Tiers and Manual Chart Review at University Sites

	Manual Chart Review Assessment and Plan (82 Patients)				Algorithm Tier 2 Charge Codes (Professional and Facility) (309 Patients)					
Laterality	*RT*	*LT*	*Agreement*[a] *(%)*	*Agreement*[a] *(%)*	*RT*	*LT*	*BL*	*Unknown*	*Agreement*[a] *(%)*	*Agreement*[b] *(%)*
Tier 1-Physician Procedure Documentation										
RT	363	1			1091	14	2	0		
LT	3	440	99.5	91.2	7	1083	4	2	98.9	89.7
BL	13	1			11	13	339	1		
Unknown	41	34			115	91	22	34		
Tier 2-Charge Codes (Professional and Facility)										
RT	395	4								
LT	6	465	98.9	98.6						
BL	14	0								
Unknown	0	2								
Tier 5-Diagnosis Code associated with Procedure										
RT	198	70			223	82	42	2		
LT	16	165	46.2	40.6	29	173	21	0	34.2	19.3
BL	174	162			434	454	156	3		
Unknown	31	78			538	492	148	63		

BL = both eyes; Charge Codes (Facility) = procedure laterality derived from Current Procedural Terminology/Healthcare Common Procedure Coding System facility fee charge code modifier; Charge Codes (Professional) = procedure laterality derived from Current Procedural Terminology professional fee charge code modifier; Crosswalked ICD-PCS Procedure Codes = procedure laterality derived from International Classification of Diseases-Procedure Coding System charge code with intrinsic laterality specification; Diagnosis Code associated with Procedure = procedure laterality derived from International Classification of Diseases, Ninth Revision/International Classification of Diseases, Tenth Revision diagnosis code associated with the procedure; LT = left eye; RT = right eye; Unknown = unspecified, unknown, or missing laterality.

Notes: University Sites: University of Southern California Roski Eye Institute.

One visit of the same patient could have multiple records. Agreement as % was calculated as the total number of cases with laterality (including all RT, LT, and BL) in agreement divided by the total number of cases.

Agreement[a] denominator was total eyes without unknown laterality.

Agreement[b] denominator was total eyes (including unknown).

Table 7. Data Accuracy: Algorithm Performance to Ascertain Cross-Sectional and Longitudinal TW-DME Status

		Health Safety Net Sites		University Sites	
	Accuracy (%)	*ICD + CPT*	*Algorithm*	*ICD + CPT*	*Algorithm*
Eye level index visit	PPV	95.8	96.6	97.1	98.6
	NPV	32.1	35.3	38.5	53.8
	Sen	73.9	78.0	84.8	91.1
	Spe	65.4	69.2	55.6	77.8
	Agreement	73.0	77.0	81.8	89.8
	F1 score	83.4	86.3	90.5	94.7
Patient level longitudinal	PPV	96.8	96.8	97.4	99.1
	NPV	30.2	32.2	48.1	71.7
	Sen	75.4	79.0	89.2	96.0
	Spe	74.0	73.0	43.2	80.7
	Agreement	75.2	78.4	84.2	94.3
	F1 score	84.7	87.0	93.1	97.5
Eye level longitudinal	PPV	47.0	93.2	85.3	98.8
	NPV	23.2	33.3	46.9	72.6
	Sen	35.9	76.0	79.2	96.0
	Spe	60.4	76.6	38.8	90.4
	Agreement	38.5	76.1	74.8	95.4
	F1 score	40.7	83.7	82.1	97.4

CPT = Current Procedural Terminology; ICD = International Classification of Diseases; NPV = negative predictive value was defined as % of chart review-confirmed nondiabetic macular edema among all the nondiabetic macular edema cases classified by the International Classification of Diseases + Current Procedural Terminology codes or algorithm; PPV = positive predictive value was defined as % of chart review-confirmed diabetic macular edema cases among all the diabetic macular edema cases classified by the International Classification of Diseases + Current Procedural Terminology codes or algorithm; Sen = sensitivity was defined as % of true diabetic macular edema cases from chart review that could be identified by the International Classification of Diseases + Current Procedural Terminology codes or algorithm; Spe = specificity was defined as % of true non diabetic macular edema cases from chart review that could be identified by the International Classification of Diseases + Current Procedural Terminology codes or algorithm; TW-DME = treatment-warranted diabetic macular edema.

Notes: Health Safety Net Sites: Los Angeles County Department of Health Services, Los Angeles General Medical Center; University Sites: University of Southern California Roski Eye Institute.

Agreement as % was calculated as the total number of cases in agreement (DME and non-DME) divided by the total number of cases. The F1 score was calculated by taking 2 times the product of PPV and sensitivity (recall), divided by the sum of PPV and sensitivity. One visit of the same patient could have multiple records for 2 eyes.

level of detail of documentation, the patient volume of the physician, the use of ancillary staff or tools to support documentation, the specific outcome of interest, and the importance of codified data for medical payment. The level of complete and accurate coding may be higher in centers that require precise coding for reimbursement of clinical services. Future analytic studies using EHRs should consider the various sources of uncertainty specific to their research using longitudinal records. The absence of any tier of information could reduce the completeness and accuracy of the algorithm, as it relies on synthesizing information from multiple data sources. For example, the University Sites had better coverage of Tier 1 (Physician note title, 90.8% vs. 75.2%) and Tier 2 (CPT code, 97.6% vs. 49.2%) compared with the health safety net sites (Table 2). Tier 1 and Tier 2 information alone sufficed to identify 98.8% laterality of the entire cohort. As a result, less accurate Tier 5 laterality information can be avoided. Additionally, the university sites had more records per patient per year compared with health safety net sites (2.3 vs. 0.9), resulting in 2.6 times the number of records per patient within the same time window. This greater coverage, higher accuracy as well as more records contribute to a higher quality of longitudinal data, which can be leveraged to address missing and erroneous DME and laterality information based on the assumptions outlined in the Methods section, ultimately leading to generally higher performance metrics for the algorithm at university sites. Although the 2 sites use different reimbursement models, one capitated and the other based on coded charges, and have variations in data completeness, quality, and patient characteristics, the algorithm enhanced efficiency across both EHR systems.

Missing or incomplete data, along with sparse clinical notes, can create gaps in laterality assignment and impact the algorithm's accuracy. This may be particularly challenging for diseases with fewer associated procedures or medications. If procedure notes mention laterality only in the note body and omit it from the title, accuracy could also be compromised. Variations in physician documentation, including different terminologies or shorthand, complicate the process. Although we employed regular expressions of common ophthalmic terms and abbreviations, the lack of uniformity across institutions and the need to adapt to evolving medical practices highlight the necessity of standardized nomenclature. We expect that this algorithm could improve data extraction efficiency for studies of eye disease, and a similar approach may be successfully applied to other settings and phenotypes. Other institutions may encounter issues due to variation in data collection and coding practices, inconsistent terminology, differences in EMR database structures, and data

quality. So, our algorithm will require additional validation to determine if its performance is as strong in other settings with unique EHR systems.

With continued research trends toward the generation of large cohorts and diverse datasets that will continue to grow over time ("big data"),[19,25] it will become imperative to develop robust methods to handle and extract data for better disease characterization. Some of the efforts are already underway in ophthalmology research; the natural language processing technique applied to images of physician notes to extract nonproliferative DR/PDR information was found to increase the sensitivity of PDR classification compared with diagnosis codes.[17,18] In addition, deep learning models applied to available OCT and fundus photos effectively detected image laterality, retinal presence, retinal field, and gradeability of DR screening images with generalization between populations.[26–29] Thus, strategies for data extraction from multiple EHR data sources and generalizability to various patient populations will be of high importance to modern research approaches and clinical care.

Incorporating longitudinal, laterality-specified phenotypes adds significant value to both patient care and research. In clinical practice, tracking disease progression and treatment responses employing data at the individual eye level enables more precise personalized care. In research, accurate eye-specific data enhances the understanding of disease progression and improves predictive models by accounting for temporal changes. These data also drive more targeted therapeutic development, ultimately advancing ophthalmology toward precision medicine.

Our algorithm, as applied to TW-DME as a test case, has several strengths. First, our analysis utilizes a large longitudinal sample from multiple sites with a diverse patient population. In addition, to the best of our knowledge, our study is the first to utilize various EHR data sources to longitudinally assess a condition at eye level. We hope our work will encourage all stakeholders to adopt universal data standards in documentation and coding and to develop more robust data sources that will improve efficiency across different EHR systems. Of note, efforts are already being made to develop universal data standards, with the American Academy of Ophthalmology selecting Systematized Nomenclature of Medicine Clinical Terms as the preferred ontology for clinical terminology.[30] As we work toward adopting more universal data standards, our algorithm may aid in making this transition smoother for various practice settings with different EHR systems, while ensuring accurate and reliable information is extracted from the EHR.

In conclusion, more accurate and reliable extraction of laterality information as demonstrated by our algorithmic approach can help streamline public health and research initiatives, contributing to overall health care quality improvement.

Footnotes and Disclosures

Originally received: October 13, 2024.
Final revision: March 21, 2025.
Accepted: April 8, 2025.
Available online: April 16, 2025. Manuscript no. XOPS-D-24-00435.

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Presented at the 2024 Annual Meeting of the Association for Research in Vision and Ophthalmology (ARVO), May 5-9, Seattle, WA.

Disclosure(s):

All authors have completed and submitted the ICMJE disclosures form.

The authors have made the following disclosures:

B.C.T.: Physician advisory board – Bausch and Lomb, Alimera, Eyepoint, Regeneron;

Financial support was provided by the National Eye Institute and the National Institutes of Health (Award Number K23EY032985 [N.A.R., B.C.T.]), National Center for Advancing Translational Science (UL1TR001855), as well as an unrestricted departmental grant to the USC Roski Eye Institute from Research to Prevent Blindness.

Support for Open Access publication was provided by the University of Southern California.

HUMAN SUBJECTS: Human subjects were included in this study. The study (HS-21-00758) was approved by the University of Southern California Institutional Review Board. Informed consent was not required because the study used preexisting, deidentified electronic health record data and met the exemption criteria under 45 Code of Federal Regulation 46.104(d)(4), posing no more than minimal risk to patients' privacy. All aspects of the research adhered to the Health Insurance Portability and Accountability Act (HIPAA) and the principles outlined in the Declaration of Helsinki.

No animal subjects were included in this study.

Author Contributions:

Conception and design: Rao, Toy, Ameri, McKean-Cowdin, Ding

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Analysis and interpretation: Rao, Toy, Ameri, McKean-Cowdin, Lang, Ding

Obtained funding: Rao, Toy

Overall responsibility: Rao, Toy, Ameri, McKean-Cowdin, Lang, Ding

Abbreviations and Acronyms:
CPT = Current Procedural Terminology; **DME** = diabetic macular edema; **DR** = diabetic retinopathy; **EHR** = electronic health record; **ICD** = International Classification of Diseases; **IRIS** = Intelligent Research in Sight; **NPV** = negative predictive value; **PDR** = proliferative diabetic retinopathy; **PPV** = positive predictive value; **TW-DME** = treatment-warranted diabetic macular edema; **TW-DR** = treatment-warranted diabetic retinopathy; **USC** = University of Southern California.

Keywords:
Electronic health record, Treatment laterality, Diabetic macular edema, Computational phenotype.

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References

1. Javitt JC. Accuracy of coding in medicare part B claims: cataract as a case study. *Arch Ophthalmol*. 1993;111:605.
2. Bearelly S. Identification of patients with diabetic macular edema from claims data: a validation study. *Arch Ophthalmol*. 2008;126:986.
3. Quigley HA, Friedman DS, Hahn SR. Evaluation of practice patterns for the care of open-angle glaucoma compared with claims data. *Ophthalmology*. 2007;114:1599–1606.
4. Latkany P, Duggal M, Goulet J, et al. The need for validation of large administrative databases: veterans Health Administration ICD-9CM coding of exudative age-related macular degeneration and ranibizumab usage. *J Ocul Biol Dis Infor*. 2010;3:30–34.
5. Pimentel MA, Browne EN, Janardhana PM, et al. Assessment of the accuracy of using *ICD-9* codes to identify uveitis, herpes zoster ophthalmicus, scleritis, and episcleritis. *JAMA Ophthalmol*. 2016;134:1001.
6. Chiu S, Davis J, Giaconi J, et al. Variable validity of computer extracted problem lists for complications of diabetes mellitus within the VA Greater Los Angeles Health System. *Diabetes Metab Syndr Clin Res Rev*. 2017;11:S611–S615.
7. Egan C, Zhu H, Lee A, et al. The United Kingdom Diabetic Retinopathy Electronic Medical Record Users Group, Report 1: baseline characteristics and visual acuity outcomes in eyes treated with intravitreal injections of ranibizumab for diabetic macular oedema. *Br J Ophthalmol*. 2017;101:75–80.
8. Lau M, Prenner JL, Brucker AJ, VanderBeek BL. Accuracy of billing codes used in the therapeutic care of diabetic retinopathy. *JAMA Ophthalmol*. 2017;135:791.
9. Biggerstaff KS, Frankfort BJ, Orengo-Nania S, et al. Validity of code based algorithms to identify primary open angle glaucoma (POAG) in Veterans Affairs (VA) administrative databases. *Ophthalmic Epidemiol*. 2018;25:162–168.
10. Stein JD, Rahman M, Andrews C, et al. Evaluation of an algorithm for identifying ocular conditions in electronic health record data. *JAMA Ophthalmol*. 2019;137:491.
11. Cai CX, Michalak SM, Stinnett SS, et al. Effect of ICD-9 to ICD-10 transition on accuracy of codes for stage of diabetic retinopathy and related complications: results from the CODER study. *Ophthalmol Retina*. 2021;5:374–380.
12. Gange WS, Lopez J, Xu BY, et al. Incidence of proliferative diabetic retinopathy and other neovascular sequelae at 5 Years following diagnosis of type 2 diabetes. *Diabetes Care*. 2021;44(11):2518–2526.
13. Malhotra NA, Greenlee TE, Iyer AI, et al. Racial, ethnic, and insurance-based disparities upon initiation of anti–vascular endothelial growth factor therapy for diabetic macular edema in the US. *Ophthalmology*. 2021;128:1438–1447.
14. Gong D, Hall N, Elze T, et al. Temporal trends in the treatment of proliferative diabetic retinopathy. *Ophthalmol Sci*. 2021;1:100037.
15. McKay KM, Apostolopoulos N, Dahrouj M, et al. Assessing the uniformity of uveitis clinical concepts and associated *ICD-10* codes across health care systems sharing the same electronic health records system. *JAMA Ophthalmol*. 2021;139:887.
16. Hamedani AG, De Lott LB, Deveney T, Moss HE. Validity of international classification of diseases codes for identifying neuro-ophthalmic disease in large data sets: a systematic review. *J Neuroophthalmol*. 2020;40:514–519.
17. Yonamine S, Ma CJ, Alabi RO, et al. Comparison of diagnosis codes to clinical notes in classifying patients with diabetic retinopathy. *Ophthalmol Sci*. 2024;4:100564.
18. Guo Y, Yonamine S, Jian Ma C, et al. Developing and validating models to predict progression to proliferative diabetic retinopathy. *Ophthalmol Sci*. 2023;3:100276.
19. Wittenborn JS, Lee AY, Lundeen EA, et al. Validity of administrative claims and electronic health registry data from a single practice for eye health surveillance. *JAMA Ophthalmol*. 2023;141:534.
20. Borkar DS, Sobrin L, Hubbard RA, et al. Techniques for improving ophthalmic studies performed on administrative databases. *Ophthalmic Epidemiol*. 2019;26:147–149.
21. Cantrell RA, Lum F, Chia Y, et al. Treatment patterns for diabetic macular edema: an intelligent research in Sight (IRIS) registry analysis. *Ophthalmology*. 2020;127:427–429.
22. Lee CS, Blazes M, Lorch A, et al. American Academy of ophthalmology intelligent research in Sight (IRIS) registry and the IRIS registry analytic center consortium. *Ophthalmol Sci*. 2022;2:100112.
23. Christine LX, Arthur B, Suzann P, et al. Trends for proliferative diabetic retinopathy vitrectomy treatments in the iris® registry (intelligent research in sight). *Invest Ophthalmol Vis Sci*. 2022;63:2195–F0258.
24. Sloan FA. Estimates of incidence rates with longitudinal claims data. *Arch Ophthalmol*. 2003;121:1462.
25. Chen JS, Copado IA, Vallejos C, et al. Variations in electronic health record-based definitions of diabetic retinopathy cohorts. *Ophthalmol Sci*. 2024;4:100468.
26. Dai L, Wu L, Li H, et al. A deep learning system for detecting diabetic retinopathy across the disease spectrum. *Nat Commun*. 2021;12:3242.
27. Liu C, Han X, Li Z, et al. A self-adaptive deep learning method for automated eye laterality detection based on color fundus photographyParanhos A, ed. *PLoS One*. 2019;14:e0222025.
28. Nderitu P, Nunez Do Rio JM, Webster ML, et al. Automated image curation in diabetic retinopathy screening using deep learning. *Sci Rep*. 2022;12:11196.
29. Jang Y, Son J, Park KH, et al. Laterality classification of fundus images using interpretable deep neural network. *J Digit Imaging*. 2018;31:923–928.
30. Hoskins HD, Hildebrand PL, Lum F. The American Academy of ophthalmology adopts SNOMED CT as its official clinical terminology. *Ophthalmology*. 2008;115:225–226.